SIMATS ENGINEERING

ASSIGNMENT - 3

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO3: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

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Dept. of CSE

QUESTIONS: 5 **Total Marks: 50**

Annexure A

ASSIGNMENT - 3

Code of Conduct:

I Barath. S (192521359) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Total
Conceptual Understanding	35 Marks					
Accuracy of Answers	25 Marks					
Application of Knowledge	15 Marks					
Completeness of Response	15 Marks					
Clarity & Presentation	10 Marks					
Total						

RUBRICS FOR CALCULATION

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	35 Marks	Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls)	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answers	25 Marks	All answers are correct, precise, and relevant	Mostly correct with minor errors	Partially correct with noticeable errors	Mostly incorrect or irrelevant
Application of Knowledge	15 Marks	Effectively applies concepts to scenarios/questions	Applies with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15 Marks	All parts of each question are fully answered	Most parts covered with small omissions	Some parts missing	Incomplete answers
Clarity & Presentation	10 Marks	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Page Replacement – Concepts

In demand paging a page is brought into a frame only when it is referenced. A reference to a page that is not in any frame is a **page fault**; a reference to a page already in memory is a **hit**. When a fault occurs and every frame is full, a replacement algorithm chooses the victim page:

- **FIFO (First-In, First-Out):** on a page fault with all frames full, the page that has been in memory the longest is replaced.
- **LRU (Least Recently Used):** on a page fault with all frames full, the page that has not been used for the longest time is replaced.
- **Optimal:** on a page fault with all frames full, the page whose next use lies farthest in the future (or that is never used again) is replaced.

$$\textit{Page-fault rate (miss rate)} = \textit{Number of page faults} / \textit{Number of references}$$

$$\textit{Hit rate} = 1 - \textit{Miss rate}$$

In every table below the frames are initially empty, so the first references to each page are compulsory faults. Each column shows the contents of the frames after that reference.

Question 1. Page Replacement Algorithms

Question: Consider the following string of page references. Show the frame allocation for three frames for: a) FIFO b) LRU c) Optimal. List the total number of page faults and miss rate for each policy.

Answer:

The page reference string for this question was not included in the question (it was given as an image that was not attached), so the frame allocation cannot be worked out. Procedure to follow once the string is available:

1. Draw a table with one column per reference and three frame rows.
2. For each reference, mark **H** if the page is already in a frame; otherwise mark **F** and load it (into an empty frame if one exists).
3. If all three frames are full on a fault, apply the rule of the algorithm (FIFO – oldest loaded; LRU – least recently used; Optimal – farthest next use).
4. Count the faults and compute Miss rate = faults / references × 100%.

Question 2. Page Replacement Algorithms

Question: Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. How many page faults would occur for the following replacement algorithms, assuming six and seven frames? 1. LRU replacement 2. FIFO replacement 3. Optimal replacement

Answer:

The string has 20 references and 7 distinct pages (1, 2, 3, 4, 5, 6, 7), so at least 7 compulsory faults occur whatever the algorithm.

(a) Assuming 6 Frames

LRU – 6 frames

LRU (Least Recently Used): on a page fault with all frames full, the page that has not been used for the longest time is replaced.

Ref	1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
Frame 1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Frame 2		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Frame 3			3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Frame 4				4	4	4	4	4	4	4	4	4	7	7	7	7	7	7	7	7
Frame 5							5	5	5	5	5	5	5	5	5	5	5	5	5	5
Frame 6								6	6	6	6	6	6	6	6	6	6	6	6	6
Result	**F**	**F**	**F**	**F**	**H**	**H**	**F**	**F**	**H**	**H**	**H**	**H**	**F**	**H**	**H**	**H**	**H**	**H**	**H**	**H**

F = page fault, **H** = hit. Total page faults (LRU) = 7, hits = 13.

FIFO – 6 frames

FIFO (First-In, First-Out): on a page fault with all frames full, the page that has been in memory the longest is replaced.

Ref	1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
Frame 1	1	1	1	1	1	1	1	1	1	1	1	1	7	7	7	7	7	7	7	7
Frame 2		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	1	1	1	1
Frame 3			3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	2	2	2
Frame 4				4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	3	3
Frame 5							5	5	5	5	5	5	5	5	5	5	5	5	5	5
Frame 6								6	6	6	6	6	6	6	6	6	6	6	6	6
Result	**F**	**F**	**F**	**F**	**H**	**H**	**F**	**F**	**H**	**H**	**H**	**H**	**F**	**H**	**H**	**H**	**F**	**F**	**F**	**H**

F = page fault, **H** = hit. Total page faults (FIFO) = 10, hits = 10.

Optimal – 6 frames

Optimal: on a page fault with all frames full, the page whose next use lies farthest in the future (or that is never used again) is replaced.

Ref	1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
Frame 1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Frame 2		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Frame 3			3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Frame 4				4	4	4	4	4	4	4	4	4	7	7	7	7	7	7	7	7
Frame 5							5	5	5	5	5	5	5	5	5	5	5	5	5	5
Frame 6								6	6	6	6	6	6	6	6	6	6	6	6	6
Result	**F**	**F**	**F**	**F**	**H**	**H**	**F**	**F**	**H**	**H**	**H**	**H**	**F**	**H**	**H**	**H**	**H**	**H**	**H**	**H**

F = page fault, **H** = hit. Total page faults (Optimal) = 7, hits = 13.

Result summary:

Algorithm	Page Faults	Hits	Miss Rate (faults/refs)	Hit Rate
LRU	7	13	7/20 = 35%	65%
FIFO	10	10	10/20 = 50%	50%
Optimal	7	13	7/20 = 35%	65%

(b) Assuming 7 Frames

LRU – 7 frames

LRU (Least Recently Used): on a page fault with all frames full, the page that has not been used for the longest time is replaced.

Ref	1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
Frame 1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Frame 2		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Frame 3			3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Frame 4				4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
Frame 5							5	5	5	5	5	5	5	5	5	5	5	5	5	5
Frame 6								6	6	6	6	6	6	6	6	6	6	6	6	6
Frame 7													7	7	7	7	7	7	7	7
Result	**F**	**F**	**F**	**F**	**H**	**H**	**F**	**F**	**H**	**H**	**H**	**H**	**F**	**H**	**H**	**H**	**H**	**H**	**H**	**H**

F = page fault, **H** = hit. Total page faults (LRU) = 7, hits = 13.

FIFO – 7 frames

FIFO (First-In, First-Out): on a page fault with all frames full, the page that has been in memory the longest is replaced.

Ref	1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
Frame 1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Frame 2		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Frame 3			3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Frame 4				4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
Frame 5							5	5	5	5	5	5	5	5	5	5	5	5	5	5
Frame 6								6	6	6	6	6	6	6	6	6	6	6	6	6
Frame 7													7	7	7	7	7	7	7	7
Result	**F**	**F**	**F**	**F**	**H**	**H**	**F**	**F**	**H**	**H**	**H**	**H**	**F**	**H**	**H**	**H**	**H**	**H**	**H**	**H**

F = page fault, **H** = hit. Total page faults (FIFO) = 7, hits = 13.

Optimal – 7 frames

Optimal: on a page fault with all frames full, the page whose next use lies farthest in the future (or that is never used again) is replaced.

Ref	1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
Frame 1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Frame 2		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Frame 3			3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Frame 4				4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
Frame 5							5	5	5	5	5	5	5	5	5	5	5	5	5	5
Frame 6								6	6	6	6	6	6	6	6	6	6	6	6	6
Frame 7													7	7	7	7	7	7	7	7
Result	**F**	**F**	**F**	**F**	**H**	**H**	**F**	**F**	**H**	**H**	**H**	**H**	**F**	**H**	**H**	**H**	**H**	**H**	**H**	**H**

F = page fault, **H** = hit. Total page faults (Optimal) = 7, hits = 13.

Result summary:

Algorithm	Page Faults	Hits	Miss Rate (faults/refs)	Hit Rate
LRU	7	13	7/20 = 35%	65%
FIFO	7	13	7/20 = 35%	65%
Optimal	7	13	7/20 = 35%	65%

Observation:

With 7 frames every distinct page fits in memory at once, so nothing is ever replaced and all three algorithms incur only the 7 compulsory faults. With 6 frames one page must be evicted, and FIFO suffers because it evicts the oldest pages 1, 2 and 3 when page 7 arrives, even though they are about to be used again, giving 10 faults. LRU and Optimal both keep the recently and soon-to-be-used pages and need only 7 faults. FIFO's faults fell from 10 to 7 as frames increased, so there is no Belady's anomaly in this case.

Final Answer:

- 6 Frames: LRU = 7, FIFO = 10, Optimal = 7.
- 7 Frames: LRU = 7, FIFO = 7, Optimal = 7.

Question 3. Page Replacement Algorithms

Question: Consider the following page reference string: 7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1. Assuming demand paging with three frames, how many page faults would occur for: 1. LRU replacement 2. FIFO replacement 3. Optimal replacement

Answer:

Number of frames = 3; number of references = 20.

LRU – 3 frames

LRU (Least Recently Used): on a page fault with all frames full, the page that has not been used for the longest time is replaced.

Ref	**7**	**2**	**3**	**1**	**2**	**5**	**3**	**4**	**6**	**7**	**7**	**1**	**0**	**5**	**4**	**6**	**2**	**3**	**0**	**1**
Frame 1	7	7	7	1	1	1	3	3	3	7	7	7	7	5	5	5	2	2	2	1
Frame 2		2	2	2	2	2	2	4	4	4	4	1	1	1	4	4	4	3	3	3
Frame 3			3	3	3	5	5	5	6	6	6	6	0	0	0	6	6	6	0	0
Result	**F**	**F**	**F**	**F**	**H**	**F**	**F**	**F**	**F**	**F**	**H**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**F**

F = page fault, **H** = hit. Total page faults (LRU) = 18, hits = 2.

FIFO – 3 frames

FIFO (First-In, First-Out): on a page fault with all frames full, the page that has been in memory the longest is replaced.

Ref	**7**	**2**	**3**	**1**	**2**	**5**	**3**	**4**	**6**	**7**	**7**	**1**	**0**	**5**	**4**	**6**	**2**	**3**	**0**	**1**
Frame 1	7	7	7	1	1	1	1	1	6	6	6	6	0	0	0	6	6	6	0	0
Frame 2		2	2	2	2	5	5	5	5	7	7	7	7	5	5	5	2	2	2	1
Frame 3			3	3	3	3	3	4	4	4	4	1	1	1	4	4	4	3	3	3
Result	**F**	**F**	**F**	**F**	**H**	**F**	**H**	**F**	**F**	**F**	**H**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**F**

F = page fault, **H** = hit. Total page faults (FIFO) = 17, hits = 3.

Optimal – 3 frames

Optimal: on a page fault with all frames full, the page whose next use lies farthest in the future (or that is never used again) is replaced.

Ref	**7**	**2**	**3**	**1**	**2**	**5**	**3**	**4**	**6**	**7**	**7**	**1**	**0**	**5**	**4**	**6**	**2**	**3**	**0**	**1**
Frame 1	7	7	7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Frame 2		2	2	2	2	5	5	5	5	5	5	5	5	5	4	6	2	3	3	3
Frame 3			3	3	3	3	3	4	6	7	7	7	0	0	0	0	0	0	0	0
Result	**F**	**F**	**F**	**F**	**H**	**F**	**H**	**F**	**F**	**F**	**H**	**H**	**F**	**H**	**F**	**F**	**F**	**F**	**H**	**H**

F = page fault, **H** = hit. Total page faults (Optimal) = 13, hits = 7.

Result summary:

Algorithm	Page Faults	Hits	Miss Rate (faults/refs)	Hit Rate
LRU	18	2	18/20 = 90%	10%
FIFO	17	3	17/20 = 85%	15%
Optimal	13	7	13/20 = 65%	35%

Observation:

Optimal gives the fewest faults (13) and is the lower bound for this string. LRU (18) performs slightly worse than FIFO (17) here: recency of use is not a good predictor of future use for this particular string, which is possible because neither algorithm looks ahead.

Final Answer:

- LRU = 18 page faults
- FIFO = 17 page faults
- Optimal = 13 page faults

Question 4. Page Replacement Algorithms

Question: Apply the following replacement algorithms for the page-reference string: 2, 6, 9, 2, 4, 2, 1, 7, 3, 0, 5, 2, 1, 2, 9, 5, 7, 3, 8, 5. 1. FIFO 2. LRU 3. Optimal (OPT). Indicate the number of page faults for each algorithm, assuming demand paging with three frames.

Answer:

Number of frames = 3; number of references = 20.

FIFO – 3 frames

FIFO (First-In, First-Out): on a page fault with all frames full, the page that has been in memory the longest is replaced.

Ref	2	6	9	2	4	2	1	7	3	0	5	2	1	2	9	5	7	3	8	5
Frame 1	2	2	2	2	4	4	4	7	7	7	5	5	5	5	9	9	9	3	3	3
Frame 2		6	6	6	6	2	2	2	3	3	3	2	2	2	2	5	5	5	8	8
Frame 3			9	9	9	9	1	1	1	0	0	0	1	1	1	1	7	7	7	5
Result	**F**	**F**	**F**	**H**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**H**	**F**	**F**	**F**	**F**	**F**	**F**

F = page fault, **H** = hit. Total page faults (FIFO) = 18, hits = 2.

LRU – 3 frames

LRU (Least Recently Used): on a page fault with all frames full, the page that has not been used for the longest time is replaced.

Ref	2	6	9	2	4	2	1	7	3	0	5	2	1	2	9	5	7	3	8	5
Frame 1	2	2	2	2	2	2	2	2	3	3	3	2	2	2	2	2	7	7	7	5
Frame 2		6	6	6	4	4	4	7	7	7	5	5	5	5	9	9	9	3	3	3
Frame 3			9	9	9	9	1	1	1	0	0	0	1	1	1	5	5	5	8	8
Result	**F**	**F**	**F**	**H**	**F**	**H**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**H**	**F**	**F**	**F**	**F**	**F**	**F**

F = page fault, **H** = hit. Total page faults (LRU) = 17, hits = 3.

Optimal – 3 frames

Optimal: on a page fault with all frames full, the page whose next use lies farthest in the future (or that is never used again) is replaced.

Ref	2	6	9	2	4	2	1	7	3	0	5	2	1	2	9	5	7	3	8	5
Frame 1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	9	9	9	3	3	3
Frame 2		6	6	6	4	4	1	1	1	1	1	1	1	1	1	1	7	7	8	8
Frame 3			9	9	9	9	9	7	3	0	5	5	5	5	5	5	5	5	5	5
Result	F	F	F	H	F	H	F	F	F	F	F	H	H	H	F	H	F	F	F	H

F = page fault, **H** = hit. Total page faults (Optimal) = 13, hits = 7.

Result summary:

Algorithm	Page Faults	Hits	Miss Rate (faults/refs)	Hit Rate
FIFO	18	2	18/20 = 90%	10%
LRU	17	3	17/20 = 85%	15%
Optimal	13	7	13/20 = 65%	35%

Observation:

Optimal gives the fewest faults (13) and is the lower bound for this string. Here LRU (17) does slightly better than FIFO (18), because recently used pages such as 2 are used again soon and LRU keeps them while FIFO may evict them.

Final Answer:

- FIFO = 18 page faults
- LRU = 17 page faults
- Optimal = 13 page faults

Question 5. Page Replacement Algorithms

Question: Apply the following replacement algorithms for the page-reference string: 0, 6, 3, 0, 2, 6, 3, 5, 2, 4, 1, 3, 0, 6, 1, 4, 2, 3, 5, 7. Find the number of page faults assuming four frames. 1. FIFO 2. LRU 3. Optimal

Answer:

Number of frames = 4; number of references = 20.

FIFO – 4 frames

FIFO (First-In, First-Out): on a page fault with all frames full, the page that has been in memory the longest is replaced.

Ref	0	6	3	0	2	6	3	5	2	4	1	3	0	6	1	4	2	3	5	7
Frame 1	0	0	0	0	0	0	0	5	5	5	5	5	0	0	0	0	0	3	3	3
Frame 2		6	6	6	6	6	6	6	6	4	4	4	4	6	6	6	6	6	5	5
Frame 3			3	3	3	3	3	3	3	3	1	1	1	1	1	4	4	4	4	7
Frame 4					2	2	2	2	2	2	2	3	3	3	3	3	2	2	2	2
Result	F	F	F	H	F	H	H	F	H	F	F	F	F	F	H	F	F	F	F	F

F = page fault, **H** = hit. Total page faults (FIFO) = 15, hits = 5.

LRU – 4 frames

LRU (Least Recently Used): on a page fault with all frames full, the page that has not been used for the longest time is replaced.

Ref	0	6	3	0	2	6	3	5	2	4	1	3	0	6	1	4	2	3	5	7
Frame 1	0	0	0	0	0	0	0	5	5	5	5	3	3	3	3	4	4	4	4	7
Frame 2		6	6	6	6	6	6	6	6	4	4	4	4	6	6	6	6	3	3	3
Frame 3			3	3	3	3	3	3	3	3	1	1	1	1	1	1	1	1	5	5
Frame 4					2	2	2	2	2	2	2	2	0	0	0	0	2	2	2	2
Result	**F**	**F**	**F**	**H**	**F**	**H**	**H**	**F**	**H**	**F**	**F**	**F**	**F**	**F**	**H**	**F**	**F**	**F**	**F**	**F**

F = page fault, **H** = hit. Total page faults (LRU) = 15, hits = 5.

Optimal – 4 frames

Optimal: on a page fault with all frames full, the page whose next use lies farthest in the future (or that is never used again) is replaced.

Ref	0	6	3	0	2	6	3	5	2	4	1	3	0	6	1	4	2	3	5	7
Frame 1	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6	6	6	6	6	6
Frame 2		6	6	6	6	6	6	5	5	4	4	4	4	4	4	4	2	2	2	2
Frame 3			3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	5	5
Frame 4					2	2	2	2	2	2	1	1	1	1	1	1	1	1	1	7
Result	**F**	**F**	**F**	**H**	**F**	**H**	**H**	**F**	**H**	**F**	**F**	**H**	**H**	**F**	**H**	**H**	**F**	**H**	**F**	**F**

F = page fault, **H** = hit. Total page faults (Optimal) = 11, hits = 9.

Result summary:

Algorithm	Page Faults	Hits	Miss Rate (faults/refs)	Hit Rate
FIFO	15	5	15/20 = 75%	25%
LRU	15	5	15/20 = 75%	25%
Optimal	11	9	11/20 = 55%	45%

Observation:

With four frames, FIFO and LRU give the same number of faults (15), and Optimal needs only 11, which is the minimum possible for this string.

Final Answer:

- FIFO = 15 page faults
- LRU = 15 page faults
- Optimal = 11 page faults

Final Answer – Quick Revision

Question	Frames	FIFO	LRU	Optimal
1	3	Data not given	Data not given	Data not given
2(a)	6	10	7	7
2(b)	7	7	7	7
3	3	17	18	13
4	3	18	17	13
5	4	15	15	11

Exam Tip:

- FIFO – Replaces the page that entered first.
- LRU – Replaces the page that was least recently used.
- Optimal – Replaces the page that will not be used for the longest time in the future.
- Optimal never produces more faults than FIFO or LRU, but it cannot be implemented in practice because it needs knowledge of future references; it is used as a yardstick.
- FIFO can show Belady's anomaly (more frames giving more faults); LRU and Optimal cannot, because they are stack algorithms.